



SARDAR PATEL UNIVERSITY
B.C.A.(3rd SEM)– EXAMINATION OCTOBER–2023
US03CBCA52:DATA STRUCTURES- I

Date: 27-10-2023 (Friday)

Time: 02:00 PM to 05:00 PM

NOTE: Right hand figure indicates the marks of each question.

Marks: [70]

Q. 1 Multiple Choice Questions (One marks each)**[10]**

- 1 Index of an array is always an _____ value.
(A) String (B) Integer
(C) Float (D) Stack
- 2 _____ are the building blocks of program.
(A) Information (B) Data Structure
(C) Data (D) Data Types
- 3 Index is also known as _____.
(A) Subscript (B) Base
(C) Non-subscript (D) Type
- 4 The term "push" and 'pop' is related to _____.
(A) Array (B) Stack
(C) Queue (D) All of these
- 5 A stack is _____ type of data structure.
(A) Linear (B) Non-Linear
(C) Both (A) and (B) (D) None of the above
- 6 Prefix notation is also called _____.
(A) Suffix notation (B) reverse polish
(C) Polish notation (D) Both (A) and (B)
- 7 A Queue is logically a _____ type of list.
(A) First-In-First-Out (B) First-In-Last-Out
(C) Last-In-First-Out (D) Last-In-Last-Out
- 8 Whenever an element is deleted from the Queue.
(A) FRONT incremented by 1 (B) FRONT decremented by 1
(C) REAR incremented by 1 (D) REAR decremented by 1
- 9 The collection of files is known as _____.
(A) Data File (B) File
(C) Database (D) None of these
- 10 A digraph is known as _____ graph.
(A) Directed (B) Multiple
(C) Double (D) Undirected

- Q. 2 Short Questions. (Attempt any ten) [20]**
- 1 Define Data Structure.
 - 2 List out application of Data Structure.
 - 3 Draw the diagram of classification of Data structure.
 - 4 List types of notation.
 - 5 List operations carried out on a Stack. Define any one of them.
 - 6 Give postfix form for $(A + B) * C / D$.
 - 7 Define queue. List the types of Queues.
 - 8 State the limitations of Simple Queue.
 - 9 What are Deques?
 - 10 What do you mean by file organization? Why it is Required?
 - 11 Define: (1) File, (2) Database.
 - 12 Explain Transaction. Which types of transactions are performed on the file?
- Q. 3 A] Explain Classification of Data Structure in detail. [05]**
B] Define array. Explain 1-D array with declaration and initialization. [05]
- OR**
- Q. 3 A] List out operations used on Data Structure. Explain any three. [05]**
B] Explain 2-D array with declaration and initialization. [05]
- Q. 4 A] Write an algorithm for PUSH and POP operations of a Stack. [05]**
B] $A + [(B + C) + (D + E) * F] / G$. Give postfix form. [05]
- OR**
- Q. 4 A] Explain recursion in detail with example. [05]**
B] Define stack and explain notations in detail. [05]
- Q. 5 A] Write an algorithm to insert and delete an element in a simple queue. [06]**
B] Explain the Circular Queues with an example. [04]
- OR**
- Q. 5 A] Explain the Simple Queues with an example. Also explain the limitations of Simple Queues. [06]**
B] Explain the Priority Queues with example. [04]
- Q. 6 A] What is buffer? Write a short note on Single buffering and Multiple buffering. [10]**
- OR**
- Q. 6 A] Write a detail note on the structure of sequential file and processing of sequential file. [10]**
